

Muhammad Zubair Zafar

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PROFESSIONAL SUMMARY

Full Stack Engineer specializing in designing and scaling web applications and APIs using Python, JavaScript, and modern frameworks (React, Flask, Node.js, Laravel). Experienced in AI/ML integration, CI/CD automation, and cloud-based deployments with Docker. Strong focus on performance optimization, RESTful architecture, and production-grade systems supporting 7,000+ active users.

EDUCATION

Eastern Illinois University

Master of Science in Computer /Information Technology | GPA: 4.0/4.0

Charleston, IL

Aug 2024 - May 2026 (Expected)

- Fully funded scholarship recipient; awarded an additional grant for initial expenses

Government College University Faisalabad

Bachelor of Computer Science

Sahiwal, Pakistan

Oct 2019 - Sep 2023

SKILLS

Languages:	Python, JavaScript (ES6+), PHP, C, Java
Frameworks:	React.js, Node.js, Express.js, Flask, Django, Laravel, Redux, Bootstrap
Databases:	MySQL, MongoDB, SQL Server, SQLite, Database Design & Query Optimization
DevOps / Tools:	Docker, Git, CI/CD (GitHub Actions), Postman, TDD, Unit & Integration Testing
AI/ML & Data:	PyTorch, TensorFlow, scikit-learn, OpenCV, NumPy, Pandas
Concepts:	REST API Development, System Design, OOP, Data Structures & Algorithms

WORK EXPERIENCE

Eastern Illinois University

Graduate Assistant / Software Engineer

Charleston, IL

Aug 2024 - Present

- Architected and deployed scalable REST APIs serving 7,000+ active users with role-based authentication, reducing API response time by 65% through query optimization and database indexing
- Engineered REST APIs integrated with Twilio SMS and automated workflow processes, reducing manual data entry by 40% and accelerating request processing by 25%
- Manage and query the application's core database using SQL (MySQL) to support 7,000+ active users and role-based access control.
- Containerize applications using Docker and worked with sysadmins to deploy via CI/CD pipelines, ensuring reliable production environments.

WPBrigade

Backend Software Engineer

Sialkot, Pakistan

Dec 2023 - Aug 2024

- Developed and maintained 5+ production WordPress plugins with 15,000+ combined active installations, implementing custom REST API endpoints, database schema optimization, and webhook integrations
- Engineered a reusable theme framework using PHP, MySQL, and JavaScript, reducing client site deployment time by 60% and enabling 200+ non-technical users to build custom websites without code
- Mentored 3 junior developers on PHP best practices, MVC architecture, and secure coding through technical bootcamps (100+ students trained)

Etihad Aviation Group, World Headquarters

Software Engineering Intern at Digital Technology and Innovation Department

Abu Dhabi, UAE

Sep 2023 - Oct 2023

- Productionized hackathon-winning "Smart CPM Parser" using Python, Flask, and regex to validate and auto-correct cargo performance messages, reducing manual processing time by 30% and eliminating 95% of data entry errors.
- Created digital data collection application using Microsoft Power Platform (Power Apps, Power Automate) for cargo team, replacing paper-based workflows and improving data accuracy by 40%
- Engineered a real-time travel verification service by integrating the Timatic REST API; built a Python backend to process and validate visa/passport data for 200+ countries.
- Engineered Code-to-Flow tool generating UML diagrams from Python codebases using py2puml library, accelerating documentation workflow for 15+ internal projects

CarChaze

Software Engineer (MERN Stack)

Lahore, Pakistan

Oct 2022 - Jan 2023

- Developed full-stack vehicle marketplace application using MongoDB, Express.js, React.js, and Node.js, serving 2,000+ monthly active users with search, filtering, and real-time messaging features
- Engineered 10+ core REST APIs in Node.js/Express.js to serve 2,000+ monthly users, optimizing data retrieval and reducing processing time by 15%.
- Deployed applications on Firebase and Heroku, managing continuous integration and delivery processes.

RESEARCH EXPERIENCE

Eastern Illinois University

Charleston, IL, USA

Research Assistant to Professor Md Farhadur Reza

Aug 2025 - Present

- Build and train deep learning models (YOLOv8, PyTorch) for multi-class medical image classification on 80,000+ labeled MRI scans, achieving 97.2% test accuracy and 95.8% F1-score.
- Develop an end-to-end data pipeline with preprocessing, augmentation (rotation, flipping, scaling), and GPU-accelerated training on Google Colab Pro, reducing training time by 70%
- Implement evaluation metrics (accuracy, precision, recall, F1-score) and automated performance monitoring with early stopping.

HACKATHON PROJECTS

Smart Industry Hackathon - By Transatlantic AI Hackathon ([GitHub](#))

Apr 2023

Smart Safety Management System

Used: Python, OAK-D, Yolo8, Numpy, Pandas, Flask

- Developed a full-stack safety system with a Flask backend to process computer vision data and an HTML/JavaScript frontend to manage employee profiles and alerts.
- Implemented real-time profile management and updates for employees, notifying higher authorities instantly.

1AHACK4SUSTAINABILITY - Hackathon By Amadeus ([Presentation](#)) ([GitHub](#))

Oct 2022

Smart CPM Parser

Used: HTML, CSS, JavaScript, Bootstrap, Python, Flask, RegEx

- Won First Place in Etihad Green Development for "Smart CPM Parser," an MVP built with Python and Flask to validate cargo messages.
- This successful MVP led directly to a 1-month internship at Etihad Aviation Group to scale and productionize the tool.

OpenAI Hackathon - LabLab.ai ([GitHub](#))

Dec 2023

Codic

Used: Python, Whisper API, Open AI API, Django

- Implemented a voice-activated image generation app integrating OpenAI Whisper, GPT-3, and Codex APIs.
- Utilized React Native and Django to develop a user-friendly interface, enabling users to generate image sbased on spoken descriptions
- Gained hands-on experience with cutting-edge AI technology and honed skills in collaborative software development.

VOLUNTEER EXPERIENCE

- Library Student Ambassadors - Eastern Illinois University
- 3x Python Teacher, Code in Place - Stanford University
- Organizer - WordSprint Developers Hackathon
- Senior Moderator & DSA Instructor - ICodeGuru
- Business and Tech Mentor - lablab.ai

Sep 2024 - Present

2023 - 2025

Feb 2024

Aug 2023 - Present

Feb 2023 - Feb 2024